

Confinement schemas and the reach of block certificates for powers of rational numbers modulo one

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ABSTRACT. Let $p > q > 1$ be coprime integers and let $Z_{p/q}(s, s + t)$ be the set of $\xi > 0$ whose fractional parts $\{\xi(p/q)^n\}$ all lie in $[s, s + t)$. In a previous paper the author described a machine-verified corpus of *block certificates* for statements of the form $Z_{p/q}(U) = \emptyset$. This paper replaces parts of that corpus by theorems. Thirty certificates in the regime $p > q^2$ collapse into one closed-form criterion on $\varepsilon = \{(p - q)s\}$, which settles a set of positions of measure $1 - 2q^2/(p(p + q))$ at every base and for every real $\xi \neq 0$; the criterion extends to a family indexed by a depth K , which we identify with the finiteness criterion of Flatto, Lagarias and Pollington and with the parameter intervals J_b^a of Bugeaud, the depth playing the role of the denominator of the rotation number. We then extract from the soundness proof an explicit escape time for $L \leq \xi \leq X$ and prove a completeness theorem: on sets with a finite hold set the certificate verdict is forced, and an infinite hold set refuses every unranked certificate at every depth, which turns a bounded computer search into a theorem. A measure-theoretic depth-size bound, $1 \leq 2\delta(K + 2 + \log_{p/q} 2B)$, quantifies the cost of the problem of Kari and Kopra on explicit unions of measure close to one for this format. Finally, a rank-stratified certificate gives $\|\xi(3/2)^n\| < 2/7$ infinitely often for every $\xi \neq 0$, and we identify the exact ceiling of the whole family as the two Thue-Morse constants of Dubickas, neither of which it can attain. All statements are theorems of Lean 4 with no unproved assumptions beyond Lean's classical core.

1. INTRODUCTION

Throughout, $\lfloor x \rfloor$ and $\{x\}$ denote the integer and the fractional part of a real number x , and $\|x\|$ the distance from x to the nearest integer. For coprime integers $p > q > 1$ and an interval $[s, s + t)$, let

$$Z_{p/q}(s, s + t) := \{\xi > 0 : s \leq \{\xi(p/q)^n\} < s + t \text{ for all integers } n \geq 0\},$$

following Flatto, Lagarias and Pollington [FLP95, (1.4)]; Mahler [Mah68] had asked whether $Z_{3/2}(0, 1/2)$ is empty. The state of the length-1/p line is this. Flatto, Lagarias and Pollington proved $Z_{p/q}(s, s + 1/p) = \emptyset$ for a dense

Date: September 12, 2026.

Institute for Globally Distributed Open Research and Education (IGDORE)
MSC 11J71, 11K06, 37E05 .

set of positions s and conjectured it for all [FLP95, Thm. 1.4, Cor. 1.4a]; Bugeaud [Bug04, Thm. 1] proved it for a set of s of full Lebesgue measure, with an explicit criterion, and showed the exceptional set to be null, uncountable and not closed [Bug04, Thm. 3]; Dubickas [Dub09AA, Thm. 1] settled every position whenever $p < q^2$, leaving $p > q^2$ open [Dub09AA, §4], and later settled that regime at every position for every *algebraic* $\xi \neq 0$ [Dub19, Thm. 1.1]. For arbitrary real ξ and $p > q^2$ the statement is open; so is the statement at the exceptional positions of [Bug04], whose elements are transcendental by [BKLN21].

Beyond single windows the printed record concerns finite unions. Dubickas [Dub08, Cor. 1.2] excludes a closed union of total length $20/39$ at the base $3/2$, while Kari and Kopra exhibit a *nonempty* confinement union of total length $2/3$ [KK17, Cor. 4.8], prove non-constructively that empty ones of total length $1 - \varepsilon$ exist for every $\varepsilon > 0$ [KK17, Thm. 5.9], and ask for explicit constructions [KK17, Prob. 6.1].

A previous paper [Ste26] adds to it a scheme of *block certificates*: finite integer witnesses for statements of the form “no $\xi \neq 0$ has its whole orbit $\{\xi(p/q)^n\}$ inside the finite union U ”. That development, to which we refer throughout as the *corpus*, is a body of instances — some forty certificates, each settling one window, one union, one base. The present paper is what happens when one asks the corpus for theorems instead: statements quantified over all bases, over intervals of real positions, over all depths, with explicit rates, and — in the other direction — statements about what the certificate format cannot do.

1.1. Results. Five groups.

First, one closed form, with a depth (§3). Say that s admits *escape data of depth K* if the orbit of the left endpoint of the window under the two-branch map that governs confinement leaves the window after exactly K steps (Definition 3.3).

Theorem A. *Let $p > q > 1$ be coprime, let s be real and let $K \geq 0$. If s admits escape data of depth K , then $Z_{p/q}(s, s + 1/p) = \emptyset$, for every $\xi \neq 0$ and with no assumption on its arithmetic nature. In particular this holds whenever*

$$q^{K+1}(p-q) \leq p\varepsilon(p^{K+1}-q^{K+1}) \quad \text{and} \quad \varepsilon(p^{K+1}-q^{K+1}) \leq q^K(p-q),$$

where $\varepsilon := \{(p-q)s\}$. (Theorems 3.5 and 3.6.)

The mechanism is a rotation: the $K+1$ marked points cut the window into $K+1$ arcs, and the map permutes the arcs cyclically (Proposition 3.4). This is not new mathematics, and §3.3 gives the attributions: the escape condition is Theorem 3.4 of [FLP95] together with Lemma 1 of [Bug04], the rotation is the one of [Bug04, Prop. 1], and the displayed interval is $J_{K+1}^1(q/p)$ in the notation of [Bug04, Lem. 3], so that Theorem A is a machine-checked fragment of [Bug04, Thm. 1]. What the formalization adds is that the criterion is uniform in (p, q, s, K) with no computation on concrete data,

and that it is the same object as the *depth* of a block certificate — which is what lets the corpus and the theory talk to each other.

Second, the certificates of the regime $p > q^2$ are its case $K \leq 1$ (§3.2). Thirty certificates of [Ste26, Thm. 7.17] — six positions at each of five bases — collapse into one criterion on the single real parameter ε .

Theorem B. *Let $p > q > 1$ be coprime integers, let s be real and put $\varepsilon := \{(p - q)s\}$. If*

$$\varepsilon = 0, \quad \text{or} \quad q \leq p\varepsilon, \quad \text{or} \quad q^2 \leq p(p + q)\varepsilon \quad \text{and} \quad (p + q)\varepsilon \leq q,$$

then $Z_{p/q}(s, s + 1/p) = \emptyset$; indeed for every $\xi \neq 0$ the parts $\{\xi(p/q)^n\}$ leave $[s, s + 1/p)$ for infinitely many n . The excluded ε form two open intervals of total length $2q^2/(p(p + q))$. (Corollaries 3.7 and 3.8.)

Here too there is no computation on concrete data: the proof is uniform in p , q and s . Specialising $\varepsilon = 0$ gives $Z_{p/q}(0, 1/p) = \emptyset$ at every coprime base, and specialising s to the six sixths at the five bases $5/2, 7/2, 9/2, 10/3, 11/3$ returns the thirty entries. Specialising instead in s gives statements the certificate scheme cannot reach at all, namely emptiness for a *continuum* of real positions in a regime where the printed statements need ξ algebraic: for instance $Z_{5/2}(s, s + 1/5) = \emptyset$ for every real $s \in [4/105, 2/21]$ (Corollary 3.11).

Third, an escape rate (§4). Every emptiness theorem in [Ste26] is proved by contradiction from the aperiodicity lemma of Dubickas and Novikas [DN05, Lem. 2], so none of them says *when* an orbit leaves. The bound latent in that argument is extractable in the sense of [Koh08], and has the following shape.

Theorem C. *Put*

$$\mathcal{E}(A, X, L) := A + 1 + \lceil \log_q(X(p/q)^A + 1) \rceil + \lceil \log_{p/q}(q/L) \rceil.$$

Let c be a valid block certificate for U at the base p/q with funnel depth K , B blocks and no rank strata. Then every ξ with $0 < L \leq \xi \leq X$ has some $n \leq K + \mathcal{E}(B, X, L)$ with $\{\xi(p/q)^n\} \notin U$. If instead s admits escape data of depth K , then every such ξ has some $n \leq \mathcal{E}(K + 2, X, L)$ with $\{\xi(p/q)^n\} \notin [s, s + 1/p)$. (Theorem 4.4.)

Both logarithms are necessary; §4.3 exhibits the second as an entry of the corpus. Nine of the ten certificates of [Ste26] acquire an explicit rate this way (Table 1); a numeral instance reads *every $\xi \in [1, 10^6]$ leaves $[\frac{1}{6}, \frac{13}{24})$ within 36 steps*. The tenth — the ranked certificate replicating [Dub08, Cor. 1.2] — admits no such bound at all, and Remark 4.7 says why this is a property of ranks rather than a gap in the proof.

Fourth, what the scheme can and cannot decide (§§5, 6). Call a point of U a *hold point* if it starts an infinite orbit of the model relation $qy_{n+1} = py_n - s_n$ inside U , and write $\mathcal{H}_{p/q}(U)$ for the set of them.

Theorem D. *Let $p > q > 1$ be coprime.*

- (i) If $\mathcal{H}_{p/q}(U)$ is finite then no $\xi \neq 0$ has all $\{\xi(p/q)^n\} \in U$ — with no certificate in sight, and no hypothesis on how the held orbits are arranged. (Theorem 5.5.)
- (ii) Conversely, if $\mathcal{H}_{p/q}(U)$ is infinite then no valid block certificate for U without rank strata exists, at any depth and over any denominator. (Theorem 5.7.)
- (iii) A valid certificate without rank strata, with B blocks and funnel depth K , must leave a hole of measure $\delta \geq (2(K + 2 + \log_{p/q} 2B))^{-1} > 0$. (Theorem 6.3 and Corollary 6.4.)

Part (i) is the converse half of the soundness theorem of [Ste26]: on the class where a certificate can exist, the verdict is already forced. Behind it is an arithmetic fact worth stating on its own — a point of $[0, 1)$ whose denominator is coprime to q has *at most one* successor of the same kind (Theorem 5.2), so the recurrent part of the model dynamics is a function and not a relation; on the recurrent set it is multiplication by p/q in $\mathbb{Z}/D\mathbb{Z}$ (Proposition 5.4), whence the P -periodic points are exactly the rationals with denominator dividing $p^P - q^P$, the shape Lagarias [Lag90] established for the $3x + 1$ problem. Part (ii) converts bounded searches into theorems: the closed arc $[0, \frac{1}{5}] \cup [\frac{4}{5}, 1]$, recorded in [Ste26] as refused by the search program to depth 60, is refused at *every* depth (Corollary 5.9) — and, the same theory predicting that ranks are exactly the device that survives an infinite hold set, a *ranked* certificate for it exists at depth 1, so that five entries of [Ste26] gain their closed-interval companions (Theorem 5.10). Part (iii) is a negative answer to [KK17, Prob. 6.1] for this format: driving the hole to 0 costs depth or blocks like $1/\delta$.

Fifth, the frontier of the method, and its exact ceiling (§7). The longest single window certified in [Ste26] has length 0.40722...; that number belonged to the search program of [Ste26], not to the certificate family.

Theorem E. $Z_{3/2}(\frac{2}{7}, \frac{5}{7}) = \emptyset$: a window of length $\frac{3}{7} = 0.428571\dots$. Moreover no $\xi \neq 0$ has all $\{\xi(3/2)^n\}$ in the closed window $[\frac{2}{7}, \frac{5}{7}]$, so for every $\xi \neq 0$,

$$\|\xi(3/2)^n\| < \frac{2}{7} = 0.285714\dots \quad \text{for infinitely many } n.$$

(Theorem 7.1.)

This passes $\frac{31}{81} = 0.38271\dots$, the longest single window excluded in print [Dub19, Thm. 1.2], and improves the corpus's previous nearest-integer constant $\frac{1}{3}$. It does *not* pass print in the nearest-integer coordinate: [Dub06, Cor. 1] gives every $\xi \neq 0$ a limit point of $(\|\xi(3/2)^n\|)_n$ at most $(1 + T(2/3))/4 = 0.285647\dots$, where $T(z) = \prod_{m \geq 0} (1 - z^{2^m})$, and $2/7$ misses it by $6.7 \cdot 10^{-5}$. Section 7.1 explains why it never will: the periodic orbits enter the centred window in a period-doubling cascade with Thue–Morse carry words whose entry points accumulate exactly at that constant, and below it Theorem 5.8 refuses every certificate. The mirror-image constant

$(3 - T(2/3))/12 = 0.238117\dots$ of the same corollary is the ceiling of the two-interval family $[0, c) \cup [1 - c, 1)$ of [Ste26], which stops at $c = 0.238$. So both ceilings of the corpus are Dubickas's two constants, and neither is attainable, because the printed constants are the accumulation points of the method's own obstruction.

Layout. Section 2 fixes notation and recalls from [Ste26] the two objects everything rests on: the carry word and the block certificate. Section 3 proves Theorems A and B, and §4 proves Theorem C. Sections 5 and 6 prove the three parts of Theorem D, and §7 proves Theorem E and locates the ceiling. Four arguments carry the paper and reward slow reading: Proposition 3.4 (the rotation), Theorem 4.2 (the extraction), Theorem 5.2 with Theorem 5.6 (determinism, and what it does to blocks), and Theorem 6.3 (the depth-size bound). The rest specialises or bookkeeps.

2. SETUP

Fix coprime integers $p > q > 1$, a real $\xi \neq 0$ and a real shift ν . Following [Dub09AA, §2], set

$$x_n := \lfloor \xi(p/q)^n + \nu \rfloor, \quad y_n := \{\xi(p/q)^n + \nu\}, \quad s_n := qx_{n+1} - px_n,$$

and call $\mathbf{w} = s_0s_1s_2\dots$ the *carry word* of the orbit (the word is introduced without a name in [DN05, §2] and [Dub09AA, (2)]). Two facts from [Ste26] are used throughout. The first is the recursion

$$(1) \quad qy_{n+1} = py_n - s_n - (p - q)\nu,$$

which is [Dub09AA, (3)]; the second drives the whole subject.

Lemma 2.1 ([DN05, Lem. 2]; [Dub09AA, Lem. 2]). *For coprime $p > q > 1$, every $\xi \neq 0$ and every ν , the carry word $\mathbf{w}$ is not eventually periodic.*

Here *eventually periodic* means: there are n_0 and $t \geq 1$ with $s_{n+t} = s_n$ for all $n \geq n_0$. Every emptiness theorem below is proved by producing, from a confinement hypothesis, an eventually periodic carry word, which Lemma 2.1 forbids. We call that closing step the *periodicity contradiction*, and name it below rather than rederive it.

We also recall the certificate scheme of [Ste26, §7] in the form the present paper consumes. Take $\nu = 0$, so that (1) reads $qy_{n+1} = py_n - s_n$ with $s_n \in \{-q + 1, \dots, p - 1\}$, and call any sequence $(y_n)_{n \geq 0}$ in $[0, 1)$ satisfying that recursion for some integers s_n an *orbit of the model relation*. A *block certificate* c for a finite union U of rational intervals at the base p/q consists of a common denominator D , a convention flag (all intervals half-open, or all closed), a *funnel*

$$T_0 := U \supseteq T_1 \supseteq \dots \supseteq T_K =: H$$

of interval lists whose last list H is the list of *blocks*, and a partition of the blocks into *rank strata* $S_0, S_1, \dots$, possibly empty. The name records the shape: condition (F) below forces each level to catch every point of U whose

successor lies in the level before it, so the lists narrow as k grows. It is *valid* when three integer conditions hold: the funnel condition (F), which says that each piece $\{y \in J : (py - s)/q \in I\}$ with $J \in U$, $I \in T_k$ is empty or lies in a single interval of T_{k+1} ; the determinism condition (D), which says that no edge $I \xrightarrow{s} J$ between blocks raises the rank and that each block has at most one outgoing edge inside its own rank; and the admissibility condition (B), $1 < q < p$ and $\gcd(p, q) = 1$. We write $K = K(c)$ for the funnel depth and $B = B(c)$ for the number of blocks, and call c *unranked* when the list of strata is empty — in which case (D) says outright that the block graph is a partial function. The soundness theorem [Ste26, Thm. 7.2] states that a valid certificate for U forbids every $\xi \neq 0$ from keeping all of $\{\xi(p/q)^n\}$ in U .

Throughout, $|A|$ denotes Lebesgue measure. The symbol T_k always denotes a funnel level; the Thue–Morse product $T(z)$ of §7.1 is always written with its argument. The proof labels are informational, not a judgement of difficulty: *Proof* marks an argument complete as it stands, *Proof sketch* one whose idea is here and whose full form is the Lean declaration listed for it in Appendix A, and a statement with neither is justified by the sentence before or after it.

3. THE SCHEMA AND THE ROTATION

The thirty certificates of [Ste26, Thm. 7.17] sit at six positions on each of five bases with $p > q^2$, and every one of them has funnel depth one and at most two blocks. This section replaces all thirty by a single statement, in which the position enters through one real number and the funnel depth through an integer parameter K ; the certificates are then the case $K \leq 1$.

Definition 3.1. For coprime $p > q > 1$ and real s put $\theta := (p - q)s$, $k := \lfloor \theta \rfloor$ and $\varepsilon := \{\theta\}$.

The parameter ε governs the problem in [Bug04] as well: it is the quantity $\{(p - q)s\}$ of [Bug04, (2)]. Work in the window coordinate $u_n := \{\xi(p/q)^n - s\}$, i.e. take $\nu = -s$ in §2, so that confinement of ξ to $[s, s + 1/p]$ is the condition $u_n < 1/p$ for all n , and (1) reads $qu_{n+1} = pu_n + \theta - s_n$. Rescale to $v := pu \in [0, 1]$; the window then has two *branches*,

- (2) low: $p(v + \varepsilon) < q$, and then $qv' = p(v + \varepsilon)$;
- (3) high: $1 \leq v + \varepsilon$, and then $qv' = p(v + \varepsilon - 1)$,

and between them sits the *hole* $q/p - \varepsilon \leq v < 1 - \varepsilon$, from which no step is possible. Up to the affine change of variable this is the map $f_{\beta, \alpha}(x) = \{\beta x + \alpha\}$ of [FLP95, §3] and [Bug04, §3] with $\beta = p/q$, and the window condition is $f_{\beta, \alpha}^n(x) < 1/\beta$.

Lemma 3.2 (the alphabet and the branches). *Suppose $u_n < 1/p$ for all n . Then $s_n \in \{k, k + 1\}$ for all n , and in the coordinate $v_n := pu_n$ every step*

$v_n \mapsto v_{n+1}$ takes the low branch (2) or the high branch (3), the branch taken being the carry letter: low for $s_n = k$, high for $s_n = k + 1$.

Proof sketch. From $0 \leq u_n, u_{n+1} < 1/p$ and $qu_{n+1} = pu_n + \theta - s_n$ one gets $\theta - q/p < s_n < \theta + 1$, an interval of length $1 + q/p < 2$ containing at most the integers $\lfloor \theta \rfloor$ and $\lfloor \theta \rfloor + 1$; this is [Dub09AA, Thm. 3], restated around $\lfloor \theta \rfloor$. Substituting $\theta = k + \varepsilon$ and $u = v/p$ into the same recursion turns $s_n = k$ into the low branch and $s_n = k + 1$ into the high one, and $0 \leq v_{n+1} < 1$ supplies the inequality that selects each. $\square$

Definition 3.3. *Escape data of depth K for (p, q, s) is a finite sequence $w_0, \dots, w_K$ of reals with $w_0 = 0$, $w_i \geq 0$ for $i \leq K$ and $w_i < 1$ for $i < K$, such that each step $w_i \mapsto w_{i+1}$, $i < K$, satisfies (2) or (3) with $\varepsilon = \{(p-q)s\}$, and such that w_K lies in the closed hole: $q \leq p(w_K + \varepsilon)$ and $w_K + \varepsilon \leq 1$. We say s escapes at depth K if such data exist.*

The right end of the hole is taken closed, which folds in the case where the orbit of the left endpoint returns to 0; the condition says that $f_{\beta, \alpha}^K(0)$ leaves $[0, 1/\beta)$, and §3.3 identifies it in the literature.

3.1. The rank is a rotation. Fix escape data w of depth K and let $r(x) := \#\{i \leq K : w_i \leq x\}$ be the *rank* of x : the index of the arc of $[0, 1)$ cut out by the $K + 1$ marked points that contains x (the symbol ν is reserved for the orbit shift of §2). Let N_L and N_R be the numbers of $i < K$ at which the low resp. the high branch is taken, so $N_L + N_R = K$.

Proposition 3.4. *Let $v \in [0, 1)$ take a branch to v' . Then*

$r(v') = r(v) + N_R + 1$ on the low branch, $r(v') + N_L = r(v)$ on the high,
and the rank decides the branch: v is low if and only if $r(v) \leq N_L$.

Proof sketch. Each branch is the affine increasing map $v \mapsto (p/q)(v + \varepsilon)$, translated by 0 or by $-p/q$, so each is order-preserving and injective on its domain; and by Definition 3.3 the marked points are carried along, $w_i \mapsto w_{i+1}$ for $i < K$. A branch therefore takes the arcs cut out by the marked points lying in its domain to arcs of the same decomposition, so $r(v') - r(v)$ is constant along it, and it is enough to evaluate the constant once. On the low branch the image is $[p\varepsilon/q, 1)$, which lies above $w_0 = 0$ and above all N_R images of high marked points, giving $+N_R + 1$; on the high branch the image is $[0, p\varepsilon/q)$, which lies below every image of a low marked point, giving $-N_L$. Finally v is low exactly when $v < q/p - \varepsilon$, and the marked points below that threshold are precisely the N_L low ones, since w_K lies in the hole; so low is the condition $r(v) \leq N_L$. $\square$

The values of r on $[0, 1)$ are $1, \dots, K + 1$, and on them the map therefore acts as the rotation by $N_R + 1$ modulo $K + 1$, of rotation number $(N_R + 1)/(K + 1)$.

Theorem 3.5. *Let $p > q > 1$ be coprime, s real and $K \geq 0$, and suppose s escapes at depth K . Then for every $\xi \neq 0$ there is n with $\{\xi(p/q)^n - s\} \geq 1/p$; consequently $Z_{p/q}(s, s + 1/p) = \emptyset$.*

Proof sketch. Suppose that $\{\xi(p/q)^n - s\} < 1/p$ for all n , and put $v_n := p\{\xi(p/q)^n - s\}$. Each step of (v_n) takes one of the two branches, and the branch taken is the carry letter (Lemma 3.2). By Proposition 3.4 the sequence $r(v_n)$ is determined by $r(v_0)$ and takes at most $K + 1$ values, hence is eventually periodic; the branch, so also the carry letter, is a function of the rank, so the carry word is eventually periodic: the periodicity contradiction. $\square$

3.2. The closed-form family. Taking the low branch K times from $w_0 = 0$ gives, by (2), $w_i + \varepsilon = \varepsilon(p^{i+1} - q^{i+1})/(q^i(p - q))$, so the escape condition at step K becomes two inequalities in ε alone.

Theorem 3.6. *Let $p > q > 1$ be coprime, s real, $\varepsilon = \{(p - q)s\}$ and $K \geq 0$. If*

$$(4) \quad q^{K+1}(p - q) \leq p\varepsilon(p^{K+1} - q^{K+1}) \quad \text{and} \quad \varepsilon(p^{K+1} - q^{K+1}) \leq q^K(p - q),$$

then s escapes at depth K ; consequently $Z_{p/q}(s, s + 1/p) = \emptyset$.

Proof sketch. The sequence $w_0 = 0$, $w_{i+1} = (p/q)(w_i + \varepsilon)$ has the closed form above, and $w_i + \varepsilon$ increases with i ; so once $w_K + \varepsilon \leq 1$ every earlier step meets its low condition $p(w_i + \varepsilon) < q$ — which is just $w_{i+1} < 1$ — automatically. Substituting the closed form at $i = K$,

$$w_K + \varepsilon = \frac{\varepsilon(p^{K+1} - q^{K+1})}{q^K(p - q)},$$

the two conditions of Definition 3.3 at the last point, $q \leq p(w_K + \varepsilon)$ and $w_K + \varepsilon \leq 1$, become exactly the two inequalities (4). $\square$

These are the escape data whose branch word is all-low. Definition 3.3 permits any word, so (4) is sufficient for depth K and not necessary: the other words cut out the other cells of the decomposition, indexed by rotation number rather than by depth alone (§3.3).

The schema of the certificates is the case $K \leq 1$, together with the degenerate case $\varepsilon = 0$, where the second block is empty outright.

Corollary 3.7 (the certificate schema). *Call s depth-one escaping at p/q if*

$$(5) \quad \varepsilon = 0 \quad \text{or} \quad q \leq p\varepsilon \quad \text{or} \quad (q^2 \leq p(p + q)\varepsilon \quad \text{and} \quad (p + q)\varepsilon \leq q).$$

Then for every such s , every $\xi \neq 0$ and every N there is $n \geq N$ with $\{\xi(p/q)^n - s\} \geq 1/p$; in particular $Z_{p/q}(s, s + 1/p) = \emptyset$.

Proof. The second and third clauses of (5) are (4) at $K = 0$ and $K = 1$ on inspection, so Theorem 3.6 applies. If $\varepsilon = 0$ the high branch (3) requires $1 \leq v$, impossible in $[0, 1)$, so by Lemma 3.2 the letter $k + 1$ never occurs

and the carry word is constant — the periodicity contradiction again. The statement for all N follows by applying this to $\xi(p/q)^N$, and the Z form by unwinding $\{\cdot - s\}$. $\square$

Corollary 3.8. *The set of $\varepsilon \in [0, 1)$ excluded by (5) is the union of the two open intervals $(0, \frac{q^2}{p(p+q)})$ and $(\frac{q}{p+q}, \frac{q}{p})$, of total length $\frac{2q^2}{p(p+q)}$.*

This is immediate from (5), since $\frac{q}{p} - \frac{q}{p+q} = \frac{q^2}{p(p+q)}$. The interval (4) has length $q^K(p-q)^2/(p(p^{K+1}-q^{K+1}))$; for $K \geq 2$ it is new and lies inside the first interval of Corollary 3.8. So the two excluded intervals are not residual noise — they decompose further, and by §3.3 the decomposition is the same at every base.

Corollary 3.9. *Let $p > q > 1$ be coprime.*

- (i) $Z_{p/q}(s, s + 1/p) = \emptyset$ whenever $(p-q)s \in \mathbb{Z}$, and in particular $Z_{p/q}(0, 1/p) = \emptyset$. At the base $3/2$ those positions are the halves, at $10/3$ the sevenths.
- (ii) $Z_{p/q}(\frac{1}{2}, \frac{1}{2} + \frac{1}{p}) = \emptyset$ whenever $p-q$ is even (then $\varepsilon = 0$) or $2q \leq p$ (then $\varepsilon = \frac{1}{2} \geq q/p$).

The remaining case of (ii) — $p-q$ odd and $p < 2q$ — is genuinely outside the depth-one schema; its smallest instance is $(p, q) = (4, 3)$, for which [Ste26] needs a certificate of funnel depth 26.

Theorem 3.10. *At each of the bases $5/2, 7/2, 9/2, 10/3, 11/3$ and each of the six positions $s = 0, \frac{1}{6}, \frac{1}{3}, \frac{1}{2}, \frac{2}{3}, \frac{5}{6}$,*

$$Z_{p/q}(s, s + \frac{1}{p}) = \emptyset.$$

Theorem 3.10 is [Ste26, Thm. 7.17] verbatim; the point is that it is now thirty specialisations of Corollary 3.7 rather than thirty computations on certificate data, each supplying only a decomposition $(p-q)s = m + \varepsilon$ with $m \in \mathbb{Z}$ and $0 \leq \varepsilon < 1$. Specialising in the other variable instead gives a kind of statement the certificate scheme cannot produce at all, since it quantifies over a continuum of positions.

Corollary 3.11 (a continuum of positions at a base with $p > q^2$). *At the base $5/2$ put $\varepsilon = \{3s\}$. Then $Z_{5/2}(s, s + \frac{1}{5}) = \emptyset$ for every real s with*

$$\begin{aligned} \text{depth } 0 : & \quad \varepsilon \geq \frac{2}{5}, \\ \text{depth } 1 : & \quad \varepsilon \in [\frac{4}{35}, \frac{2}{7}], \\ \text{depth } 2 : & \quad \varepsilon \in [\frac{8}{195}, \frac{4}{39}], \\ \text{depth } 3 : & \quad \varepsilon \in [\frac{16}{1015}, \frac{8}{203}]; \end{aligned}$$

in particular for every real $s \in [\frac{4}{105}, \frac{2}{21}]$ and every real $s \in [\frac{8}{585}, \frac{4}{117}]$.

Here $p = 5 > q^2 = 4$, so [Dub09AA, §4] leaves the statement open and [Dub19, Thm. 1.1] settles it only for algebraic ξ ; the positions are arbitrary

reals, and so is ξ . The depth-two row is 54% of the residual interval $(0, \frac{4}{35})$ left by depth one at this base, and the family (4) over all K covers about 87% of it.

3.3. Provenance. The mathematics of this section is not new; this subsection states the correspondence precisely and is used nowhere below, so a reader tracking the arguments can skip it. Definition 3.3 is the finiteness criterion [FLP95, Thm. 3.4] — for N least with $f_{\beta, \alpha}^N(0) \geq 1/\beta$ the survivor set has exactly N elements, cyclically permuted — together with the returning case [Bug04, Lem. 1], which [Bug04, Lem. 2] turns into an equivalence. Proposition 3.4 is that cyclic permutation, and [Bug04, Prop. 1], resting on [Bug93] and [BC99], identifies it as the rotation number of a *contracted rotation* $T_{\gamma, \alpha}(x) = \{\gamma x + \alpha\}$ [BKLN21, Def. 1]. The interval (4) is $J_{K+1}^1(q/p)$ in the notation of [Bug04, Lem. 3], the other rotation numbers a/b giving the other cells $J_b^a(q/p)$ — which is why the decomposition of the residual intervals is base-independent: the cells are indexed by rotation numbers, not by the base. Assembled over all of them, [Bug04, Thm. 1] states that $Z_{p/q}(s, s + 1/p) = \emptyset$ for a set of s of full Lebesgue measure and [Bug04, Thm. 3] that the exceptional set is null, uncountable and not closed, so Theorems 3.5 and 3.6 are a machine-checked fragment of that theorem, not an improvement of it.¹ For $p < q^2$ the emptiness is in any case [Dub09AA, Thm. 1] at every s ; what is new there is only which positions are settled at which depth. What the formalization adds is that the criterion is proved uniformly in (p, q, s, K) , with no computation on concrete data, and that Definition 3.3 is *identified* with the funnel depth of a block certificate — which is what lets §4 treat the two on the same footing.

4. AN ESCAPE RATE

Every emptiness statement so far, and every one in [Ste26], has the same logical shape: the periodicity contradiction of §2. Such a proof says nothing about *when* a given ξ leaves the set. The bound is nevertheless latent in it, in the sense of the metatheorems of [Koh08, §2], and this section extracts it. The extraction has three steps: a combinatorial one, which turns confinement over a long enough range into an exact period of the carry word; an arithmetic ladder; and a dichotomy that contradicts the period.

4.1. The ladder and the dichotomy. For $P \geq 1$ write $M_n := x_{n+P} - x_n$ for the increment of the integer part over one period.

¹Two sets of ε remain: $\varepsilon = 0$, covered by Corollary 3.9(i), and those whose marked orbit never escapes — by [Bug04, Thm. 1] exactly the $\varepsilon = \frac{p-q}{p} \sum_{k \geq 1} \varepsilon_{-k}(\tau)(q/p)^k$ with τ irrational, $\varepsilon_k(\tau) := \lfloor k\tau \rfloor - \lfloor (k-1)\tau \rfloor$ being the characteristic Sturmian sequence of slope τ [Bug04, §2] (that subscripted symbol is Bugeaud’s, unrelated to the ε of Definition 3.1). They form a null uncountable set of transcendental numbers [BKLN21], and emptiness there is open: “we have not been able to determine whether $Z_{p/q}(s, s + 1/p)$ is empty for all values of s ” [Bug04, §2].

Definition 4.1. For $A \in \mathbb{N}$ and reals $0 < L \leq X$ put

$$\mathcal{E}(A, X, L) := A + 1 + \left\lceil \log_q(X(p/q)^A + 1) \right\rceil + \left\lceil \log_{p/q}(q/L) \right\rceil.$$

Theorem 4.2. Let $p > q > 1$ be coprime and $0 < L \leq |\xi| \leq X$. If $1 \leq P$, $t + P \leq A$ and $M \geq \mathcal{E}(A, X, L)$, then the carry word cannot satisfy $s_{n+P} = s_n$ for all n with $t \leq n$ and $n + P + 1 \leq M$.

Proof sketch. Suppose it did. Subtracting the identity $qx_{n+1} = px_n + s_n$ at n from the same identity at $n + P$ gives $qM_{n+1} = pM_n$; iterating, $q^m M_{t+m} = p^m M_t$ for every m with $t + m + P \leq M$, whence $q^m \mid M_t$ since $\gcd(q^m, p^m) = 1$. This is the step “ $q^\ell \mid M$ ” of [Dub09AA, §3], made quantitative: the exponent is exactly the length of the range on which periodicity is known. Taking $m := M - t - P$, the ladder yields the dichotomy of [Dub09AA, Lem. 2],

$$(6) \quad q^{M-t-P} \leq |\xi|(p/q)^{t+P} + 1 \quad \text{or} \quad |\xi|(p/q)^{M-P}((p/q) - 1) < 1 :$$

if $M_t \neq 0$ then $q^m \leq |M_t| = |x_{t+P} - x_t| < |\xi|(p/q)^{t+P} + 1$, because x_n differs from $\xi(p/q)^n + \nu$ by a quantity in $[0, 1)$; and if $M_t = 0$ then the ladder forces $M_{M-P} = 0$, so $x_M = x_{M-P}$, which compared with $x_n = \xi(p/q)^n + \nu - y_n$ at the two indices gives the second alternative. Both now fail, for different reasons: with $t + P \leq A$ and $|\xi| \leq X$ the first bounds M by $t + P + \lceil \log_q(X(p/q)^A + 1) \rceil$, and with $|\xi| \geq L$ and $(p/q) - 1 \geq 1/q$ the second bounds it by $P + \lceil \log_{p/q}(q/L) \rceil$. Either way $M < \mathcal{E}(A, X, L)$, against the hypothesis. $\square$

The number A is the *combinatorial budget*: an index by which confinement forces the carry word to repeat. Two front ends supply it.

Proposition 4.3. (i) If s escapes at depth K and $\{\xi(p/q)^n - s\} < 1/p$ for all $n \leq N$, then there are t and $P \geq 1$ with $t + P \leq K + 2$ such that the carry word of the orbit with shift $\nu = -s$ satisfies $s_{n+P} = s_n$ for all n with $t \leq n$ and $n + P + 1 \leq N$.
 (ii) If c is a valid unranked block certificate for U with funnel depth K and B blocks, if $B \leq N$, and if $\{\xi(p/q)^n\} \in U$ for all $n \leq N + K$, then the same holds with $t + P \leq B$.

Proof sketch. For (i) the rank sequence of Proposition 3.4 is deterministic with values in a set of $K + 1$ elements, so two of the first $K + 2$ values coincide and the sequence — hence the carry word, which is a function of the rank — repeats from there. For (ii) the funnel condition pushes the orbit into the blocks within K steps, and for an unranked certificate the block graph is a partial function, so the block itinerary is determined by its first block and repeats within B steps, carrying the letters read on the edges with it. $\square$

4.2. The bound.

base	certified set	K	B	β	escape bound for $1 \leq \xi \leq X$
3/2	$[\frac{1}{6}, \frac{13}{24})$	8	3	2	$14 + \alpha, 2^\alpha \geq X(3/2)^3 + 1$
3/2	$[\frac{961}{3600}, \frac{2427}{3600})$	13	2	2	$18 + \alpha, 2^\alpha \geq X(3/2)^2 + 1$
3/2	union of total length $\frac{7}{12}$	7	1	2	$11 + \alpha, 2^\alpha \geq X(3/2) + 1$
3/2	union of total length $\frac{2}{3}$	9	12	2	$24 + \alpha, 2^\alpha \geq X(3/2)^{12} + 1$
3/2	union of total length $\frac{25}{36}$	11	17	2	$31 + \alpha, 2^\alpha \geq X(3/2)^{17} + 1$
3/2	union of total length $\frac{17}{24}$	17	100	2	$120 + \alpha, 2^\alpha \geq X(3/2)^{100} + 1$
3/2	$[0, \frac{1}{5}) \cup [\frac{4}{5}, 1)$	1	2	2	$6 + \alpha, 2^\alpha \geq X(3/2)^2 + 1$
4/3	$[\frac{1}{3}, \frac{5}{8})$	5	2	4	$12 + \alpha, 3^\alpha \geq X(4/3)^2 + 1$
5/2	$[\frac{1}{5}, \frac{2}{5})$	1	1	1	$4 + \alpha, 2^\alpha \geq X(5/2) + 1$

TABLE 1. The nine unranked certificates of [Ste26], made effective by Theorem 4.4. Each row asserts: for every $\xi \in [1, X]$ there is n within the stated bound with $\{\xi(p/q)^n\}$ outside the certified set. Here K and B are the certificate's own funnel depth and block count, and β is the least integer with $q \leq (p/q)^\beta$.

Theorem 4.4 (the escape bound). *Let $p > q > 1$ be coprime. In each of the two cases below, every ξ with $0 < L \leq \xi \leq X$ has some*

$$n \leq \delta + \mathcal{E}(A, X, L) \quad \text{with} \quad \{\xi(p/q)^n\} \notin S,$$

where the offset δ , the combinatorial budget A and the set S are

	δ	A	S
s escaping at depth K	0	$K + 2$	$[s, s + \frac{1}{p})$
c unranked for U , funnel depth K , B blocks	K	B	U .

In the second case U may be replaced by any set that the certificate covers.

Both rows follow from Theorem 4.2 and Proposition 4.3, whose two parts supply the budget A ; the offset is the difference between their hypotheses. In row one the rank of v_n is read off v_n alone, so confinement up to N is all the pigeonhole needs and $\delta = 0$. In row two the funnel condition looks *forward* — a point lies in a block only if the orbit stays in U for K further steps — so confinement must be assumed out to $N + K$ in order to run the pigeonhole on $[0, N]$, and that surplus is the offset $\delta = K$. For a certificate entry one wants the two logarithms replaced by numerals, and it suffices to exhibit exponents dominating them: if $X(p/q)^B + 1 \leq q^\alpha$ and $q \leq (p/q)^\beta$, then for $\xi \in [1, X]$ the bound reads $n \leq K + B + 1 + \alpha + \beta$. Table 1 lists the result for the certificates of [Ste26].

Corollary 4.5. *Every ξ with $1 \leq \xi \leq 10^6$ has some $n \leq 36$ with $\{\xi(3/2)^n\} \notin [\frac{1}{6}, \frac{13}{24})$.*

Proof. Row one of Table 1 with $\alpha = 22$, since $10^6 \cdot (3/2)^3 + 1 < 2^{22}$. $\square$

4.3. Both logarithms are necessary. The plain reading of Theorem 4.4 would ask for a bound in X alone — an escape time uniform over $0 < \xi \leq X$. There is none, and the counterexample is an entry of the table.

Remark 4.6. The seventh row certifies $[0, \frac{1}{5}) \cup [\frac{4}{5}, 1)$ at the base $3/2$. For $0 < \xi < \frac{1}{5}$ one has $\lfloor \xi(3/2)^n \rfloor = 0$ and $\{\xi(3/2)^n\} = \xi(3/2)^n$ as long as the latter is below $\frac{1}{5}$, so the orbit stays inside the certified set for exactly $\lceil \log_{3/2}(1/(5\xi)) \rceil$ steps, which is unbounded as $\xi \rightarrow 0$. Hence the term $\lceil \log_{p/q}(q/L) \rceil$ of Definition 4.1 cannot be dropped. The other logarithm converts the first alternative of (6) into a bound on M , and grows with X for the same reason that $|M_t|$ does.

Remark 4.7 (ranked certificates admit no bound). Theorem 4.4 and Proposition 4.3(ii) require the certificate to be unranked. This is not an artefact of the proof. For a ranked certificate the validity condition only asks that each block have at most one outgoing edge *within its own rank*, and the soundness argument uses that a non-increasing rank is eventually constant — a statement with no rate. A block may carry both an equal-rank edge and a lower-rank one, so a model orbit can follow an equal-rank cycle for arbitrarily long before dropping, and the carry word becomes periodic only after the last drop. No bound in the certificate’s data alone exists. The one certificate of [Ste26] this excludes is the ranked entry replicating [Dub08, Cor. 1.2], whose hold set has transients feeding two cycles — which is why ranks were introduced there in the first place.

5. WHAT THE SCHEME CAN DECIDE

The soundness theorem [Ste26, Thm. 7.2] says a certificate forces the verdict. This section proves the converse half — the class on which a certificate can exist already forces the verdict, without one — and then the exact obstruction to the scheme, which turns bounded computer searches into theorems.

Definition 5.1. The *hold set* $\mathcal{H}_{p/q}(U)$ of a set $U \subseteq \mathbb{R}$ is the set of z that start an orbit of the recursion lying entirely in U : there are reals $(y_n)_{n \geq 0}$ and integers (s_n) with $y_0 = z$, $y_n \in U$ for all n , and $qy_{n+1} = py_n - s_n$ for all n . (For $U \subseteq [0, 1)$, which is the only case used below, these are orbits of the model relation.)

The hold set is the survivor set of the interval map at the level of the model: a certificate’s funnel pushes it into the blocks, and a real orbit confined to U runs inside it. It is generally nonempty even for a certified U ; the certificate rules out a *real number* whose fractional parts follow the model forever, not the model orbits themselves.

5.1. The recurrent dynamics is a function. Say a real y has denominator coprime to q if $Dy \in \mathbb{Z}$ for some $D \geq 1$ with $\gcd(D, q) = 1$.

Theorem 5.2 (determinism). *Let u be real and $v, v' \in [0, 1)$, all three with denominators coprime to q , and suppose $qv = pu - s$ and $qv' = pu - s'$ for some integers s, s' . Then $v = v'$.*

Proof. Put all three over a common denominator D coprime to q , say $u = a/D$, $v = b/D$, $v' = c/D$ with $a, b, c \in \mathbb{Z}$. The two recursions give $q(b - c) = (s' - s)D$, so $D \mid q(b - c)$ and hence $D \mid b - c$; but $b, c \in [0, D)$, so $b = c$. $\square$

The arithmetic content is that among the q carries admissible at a point of $[0, 1)$ — which are q consecutive integers — exactly one keeps the denominator coprime to q . So the recurrent part of the model dynamics is a map, not a relation. The recurrent points are the rational cycles, and their denominators are the ones Lagarias identified for the $3x + 1$ problem.

Proposition 5.3. *Let $p > q > 1$ be coprime and let (y_n) be reals satisfying $qy_{n+1} = py_n - s_n$ for integers s_n , with $y_a = y_b$ for some $a \leq b$. Then $(p^{b-a} - q^{b-a})y_m \in \mathbb{Z}$ for every m with $a \leq m \leq b$. In particular, if $y_P = y_0$ then $(p^P - q^P)y_0 \in \mathbb{Z}$, and $\gcd(p^P - q^P, q) = 1$ for $P \geq 1$.*

Proof. Iterating the recursion gives $q^\ell y_{n+\ell} = p^\ell y_n - c$ with $c \in \mathbb{Z}$, for every n and ℓ . Apply this on $[a, m]$ and on $[m, b]$, say $q^{m-a}y_m = p^{m-a}y_a - c_1$ and $q^{b-m}y_b = p^{b-m}y_m - c_2$, and eliminate $y_a = y_b$:

$$(p^{b-a} - q^{b-a})y_m = p^{m-a}c_2 + q^{b-m}c_1 \in \mathbb{Z}.$$

The special case is $a = m = 0$, $b = P$. For the last clause, modulo any prime dividing q one has $p^P - q^P \equiv p^P$, which $\gcd(p, q) = 1$ keeps nonzero. $\square$

This is the shape of [Lag90, Thm. 2.1] at a general coprime base; [Ste26] had it at $3/2$ only. It has a converse, and the converse comes with the closed form of the map.

Proposition 5.4. *Let $D \geq 1$ with $\gcd(D, pq) = 1$ and $0 \leq a < D$. Then the orbit of a/D under the recurrent map is*

$$\frac{a}{D} \mapsto \frac{(paq^{-1} \bmod D)}{D},$$

that is, multiplication by the unit p/q of $\mathbb{Z}/D\mathbb{Z}$. Consequently

- (i) the orbit is purely periodic — there are no transients in the recurrent set;
- (ii) if $D \mid p^P - q^P$ then the orbit has period dividing P .

Proof sketch. For the formula, q is invertible modulo D , and Theorem 5.2 says the successor with denominator coprime to q is unique, so it is the one the recursion forces: $v \equiv pu/q$ in $\mathbb{Z}/D\mathbb{Z}$. For (i), multiplication by a unit of $\mathbb{Z}/D\mathbb{Z}$ is a bijection, so the orbit returns to its starting point after $\text{ord}(p/q)$ steps. For (ii), $D \mid p^P - q^P$ makes $(p/q)^P = 1$ in $\mathbb{Z}/D\mathbb{Z}$. $\square$

With Proposition 5.3 this pins the periodic points exactly: for coprime $p > q > 1$ and $P \geq 1$ the P -periodic points of the model relation in $[0, 1)$ are

$$\left\{ \frac{A}{p^P - q^P} : 0 \leq A < p^P - q^P \right\}$$

— the denominator is forced, and every numerator occurs.

5.2. Completeness.

Theorem 5.5 (completeness on the finite-hold-set class). *Let $p > q > 1$ be coprime and let $U \subseteq \mathbb{R}$ have a finite hold set. Then no $\xi \neq 0$ has $\{\xi(p/q)^n\} \in U$ for every n .*

Proof. If some $\xi \neq 0$ did, its own orbit would lie in $\mathcal{H}_{p/q}(U)$, hence have finite range, so some value recurs infinitely often; from its first occurrence n_0 onwards every point of the orbit lies on a loop, hence has denominator $p^P - q^P$ for some $P \geq 1$ by Proposition 5.3 and so denominator coprime to q . On that class the relation is a function (Theorem 5.2), so equal values at two times force equal values ever after: the orbit, and with it the carry word, is periodic from n_0 on, which is the periodicity contradiction. $\square$

So on the class where a certificate can exist, the verdict is *empty* and it is correct, with no certificate in sight and with no hypothesis on how the held orbits are arranged, since Theorem 5.2 makes the branching hypothesis one would otherwise have to assume automatic. What remains unformalized on this side is the production of the certificate *data* from a finite hold set; that is combinatorics of interval lists rather than dynamics, and the search program of [Ste26] does it.

5.3. The obstruction.

Theorem 5.6. *Let c be a valid unranked certificate for U and let $(y_n), (y'_n)$ be two orbits of the model relation confined to U . If y_0 and y'_0 lie in the same block of c , then $y_0 = y'_0$.*

Proof sketch. For an unranked certificate the block graph is a partial function, so the block itinerary and the carry word of a confined orbit are determined by its first block; hence the two orbits carry the same carry word. Now solve (1) the other way round, $y_n = (qy_{n+1} + s_n)/p$: iterating gives $y_n = \sum_{i < m} q^i s_{n+i}/p^{i+1} + (q/p)^m y_{n+m}$, and the tail vanishes because $q < p$ and y stays in $[0, 1)$. So an orbit is determined by its carry word,

$$y_n = \sum_{i \geq 0} \frac{q^i s_{n+i}}{p^{i+1}},$$

and the two orbits coincide. $\square$

By pigeonhole, a valid unranked certificate for U therefore has at least $|\mathcal{H}_{p/q}(U)|$ blocks; and since a certificate has finitely many blocks whatever its depth, an infinite hold set refutes the unranked scheme outright.

Theorem 5.7. *If $\mathcal{H}_{p/q}(U)$ is infinite then no valid unranked certificate for U exists — at any funnel depth and over any denominator.*

Theorem 5.8. *Let c be an unranked certificate at the base p/q . If infinitely many distinct points of $[0, 1)$ with denominator coprime to pq keep their whole orbit inside the certified set, then c is invalid. Equivalently: the removed set $[0, 1) \setminus U$ must meet all but finitely many cycles of the model relation.*

Theorem 5.8 is Theorem 5.7 combined with Proposition 5.4, and it says that a certificate’s hole is a *transversal* of the cycle set. The cycles are dense in $[0, 1)$, so this is a real constraint.

5.4. Closed endpoints: a search refusal becomes a theorem. The half-open two-interval union $[0, \frac{1}{5}) \cup [\frac{4}{5}, 1)$ carries a certificate of funnel depth 1 [Ste26, §7.4]. With both endpoints closed, [Ste26] records that the search program finds no certificate to depth 60. Both halves of that story are now theorems, and they say different things.

Corollary 5.9. *No unranked certificate for a set containing $[0, \frac{1}{5}] \cup [\frac{4}{5}, 1]$ is valid, at any depth and over any denominator.*

Proof sketch. At the base $3/2$ the branch of the model relation with carry 0 carries $\frac{1}{5}(\frac{2}{3})^{k+1}$ to $\frac{1}{5}(\frac{2}{3})^k$, so the chain $\frac{1}{5}(\frac{2}{3})^k$, $k \geq 0$, runs upwards to $\frac{1}{5}$; from there the carries -1 and 2 alternate around the 2-cycle $\frac{1}{5} \leftrightarrow \frac{4}{5}$. Every point of the chain therefore starts an orbit that stays in $[0, \frac{1}{5}] \cup [\frac{4}{5}, 1]$ for ever, so the hold set is infinite and Theorem 5.7 applies. $\square$

So the search’s refusal was a refusal of the unranked search, not of the set. The theory predicts the complement: rank strata are exactly the device that survives an infinite hold set, and a ranked certificate does exist — at depth one.

Theorem 5.10. *For every $\xi \neq 0$ there is n with $\{\xi(p/q)^n\}$ outside the closed set, in each of the following five cases:*

$$\begin{array}{lll} [0, \frac{1}{5}] \cup [\frac{4}{5}, 1], & [\frac{1}{6}, \frac{13}{24}], & [\frac{961}{3600}, \frac{2427}{3600}] & \text{at the base } 3/2, \\ [\frac{1}{3}, \frac{5}{8}] & & & \text{at } 4/3, \\ [\frac{1}{5}, \frac{2}{5}] & & & \text{at } 5/2. \end{array}$$

The first is certified at funnel depth 1 with 4 blocks in 2 rank strata: the 2-cycle $\{\frac{1}{5}, \frac{4}{5}\}$ at rank 0 and the two large intervals at rank 1, the backward chain $\frac{1}{5}(\frac{2}{3})^k$ living in the rank-1 block at 0 — exactly the configuration Corollary 5.9 shows an unranked certificate cannot have. The remaining four are the closed companions of four half-open entries of [Ste26]: closing an endpoint adds transients that only a rank stratification separates.

Remark 5.11 (the obstruction is not what it looks like). It is tempting to guess that the obstruction to the closed convention is an endpoint-touching

cycle. That is false: the closed arc $[0, \frac{1}{5}] \cup [\frac{4}{5}, 1]$ has the endpoint cycle $\frac{1}{5} \leftrightarrow \frac{4}{5}$ and a certificate all the same, and so does the closed union of [Dub08, Cor. 1.2]. The true obstruction to the unranked scheme is an infinite hold set — which is what an endpoint cycle usually produces, by admitting a backward chain accumulating on it.

Remark 5.12. The first entry of Theorem 5.10 is subsumed as a statement: [Ste26, Thm. 7.14] already gives $\|\xi(3/2)^n\| \geq 0.238$ for some n , and $\|y\| \leq \frac{1}{5}$ implies $\|y\| < 0.238$. It is kept for the certificate, not for the statement.

6. THE DEPTH-SIZE BOUND

The certified unions have been getting larger — $\frac{20}{39}$ in [Dub08, Cor. 1.2], then $\frac{25}{36}$ and $\frac{17}{24}$ in [Ste26] — and [KK17, Prob. 6.1] asks for a family whose measure tends to 1. This section prices that request for the certificate format: a certificate cannot certify a set of measure close to 1 without paying, in funnel depth or in blocks, like $1/\delta$.

Write $\delta := |[0, 1] \setminus U|$ for the measure of the hole. Let $f^{-1}(S)$ denote the set of points of $[0, 1]$ having a successor in S under the model relation, and let

$$T_0 := U, \quad T_{k+1} := U \cap f^{-1}(T_k)$$

be the *exact funnel* of U : the points of U with a chain of k successors inside U . A certificate's own levels are supersets of these.

Proposition 6.1 (expansion). *For measurable $S \subseteq [0, 1]$ one has $|S| \leq |f^{-1}(S)|$. Consequently $|T_k| \geq 1 - (k + 1)\delta$ for every k .*

Proof sketch. Factor $f^{-1} = \text{slice}_p \circ \text{fold}_q$, where $\text{fold}_q(S) = \{\{qv\} : v \in S\}$ and $\text{slice}_n(A) = \{y \in [0, 1] : \{ny\} \in A\}$, both written as finite unions of affine preimages. Unfolding preserves measure exactly, since the n pieces of slice_n are disjoint translates scaled by $1/n$; folding cannot shrink, since $S \subseteq \text{slice}_q(\text{fold}_q S)$. The second statement is then an induction from $|T_0| = 1 - \delta$:

$$|T_{k+1}| = |U \cap f^{-1}(T_k)| \geq |f^{-1}(T_k)| - \delta \geq |T_k| - \delta.$$

□

Proposition 6.2 (contraction). *Let c be a valid unranked certificate for U with funnel depth K and B blocks. Then two orbits confined to U for $K + j$ steps whose starting points lie in the same block satisfy $|y_0 - y'_0| \leq (q/p)^j$, and consequently*

$$|T_{K+j}| \leq B(q/p)^j \quad \text{for every } j \geq 0.$$

Proof sketch. This is the finite-range form of Theorem 5.6: an unranked block graph forces both orbits to carry the same carry word for j steps, and two orbits with the same word satisfy $q^j|y_j - y'_j| = p^j|y_0 - y'_0|$, which $[0, 1]$ bounds. Each of the B blocks therefore holds a subset of T_{K+j} of diameter at most $(q/p)^j$. □

The two propositions pull against each other, and comparing them at depth $K + j$ and optimising in j gives the bound.

Theorem 6.3. *Let c be a valid unranked block certificate at the base p/q with funnel depth K and $B \geq 1$ blocks, certifying the union U . Then*

$$1 \leq 2\delta \left(K + 2 + \log_{p/q}(2B) \right), \quad \text{i.e.} \quad \frac{1}{2\delta} \leq K + 2 + \log_{p/q}(2B).$$

Proof. Take $k = K + j$ in Proposition 6.1 and compare with Proposition 6.2:

$$1 - (K + j + 1)\delta \leq |T_{K+j}| \leq B(q/p)^j.$$

Choose $j := \lceil \log_{p/q}(2B) \rceil$, so that $(p/q)^j \geq 2B$ and hence $B(q/p)^j \leq \frac{1}{2}$. Then $1 - (K + j + 1)\delta \leq \frac{1}{2}$, that is $1 \leq 2\delta(K + j + 1)$; and $j \leq \log_{p/q}(2B) + 1$, so $K + j + 1 \leq K + 2 + \log_{p/q}(2B)$. The $+2$ of the statement is exactly this: one from the level count $k + 1$, one from the ceiling. $\square$

Corollary 6.4. *A valid unranked certificate leaves a hole of positive measure; in fact $\delta \geq (2(K + 2 + \log_{p/q} 2B))^{-1}$, and the certified set has measure < 1 .*

This sharpens the elementary observation of [Ste26, Rem. 7.3] that a certificate cannot cover all of $[0, 1)$: it cannot cover a set of full measure either, and the deficiency is quantified.

Example 6.5. The certificate of [Ste26] for the union of total length $\frac{25}{36}$ has $K = 11$ and $B = 17$, so Theorem 6.3 gives $\delta \geq (2(13 + \log_{3/2} 34))^{-1} \geq \frac{1}{44}$. The true hole is $\frac{11}{36} = 0.305\dots$, so at this size the bound is slack by a factor of about thirteen.

Theorem 6.3 is not a competitive estimate for one set. What it asserts is that the slack *closes* as $\delta \rightarrow 0$, and that is what settles [KK17, Prob. 6.1] for this format. Concretely, a family of unions with $\delta_P \leq C(q/p)^{cP}$ — the natural rate to hope for — would force $K_P + \log_{p/q} B_P \gtrsim C^{-1}(p/q)^{cP}$, so the funnel depth grows exponentially in P ; and since the funnel's common denominator is a multiple of p^K , the certificate *data* would grow doubly exponentially. A parametric certificate of bounded shape, valid for every P , does not exist in this format.

Remark 6.6. Measurement, not theorem, and nothing later depends on it: for completeness we record what the search itself reaches — restarting the greedy, interval-by-interval search of [Ste26] from its 240-interval record produces unions of total length $\frac{179}{240}$, $\frac{3}{4}$ and $\frac{181}{240} = 0.754167\dots$, each verdict rechecked by an independent exact funnel computation. At depth 29 with 2801 blocks these are far too large to verify inside the formalization, so they are search output only. They are records on the curve of [KK17, Prob. 6.1], not a solution of it. Theorem 5.8 is visible in them: of the 531 292 orbits of period at most 14 at the base $3/2$, each record keeps one or two — the fixed point 0, and at two of them the 3-cycle $\frac{2}{19} \rightarrow \frac{3}{19} \rightarrow \frac{14}{19}$, whose carry word is 0, $-1, 2$ — so the removed set is a hitting set for the whole census.

7. THE RANKED FRONTIER AND ITS EXACT CEILING

The development [Ste26] records a frontier for single windows at the base $3/2$: length $L^* = \frac{1466}{3600} = 0.407222\dots$, at fourteen positions, and nothing at all at $\frac{1467}{3600}$. That number belonged to the search program, not to the certificate family. The programs of [Ste26] merge the surviving components of each funnel level into interval hulls and then require out-degree at most one; the *rank-stratified* criterion — sound by the same soundness theorem, and already used there for two entries — certifies the same row at 213 of its 236 positions instead of fourteen, and climbs past L^* .

Theorem 7.1. $Z_{3/2}(\frac{2}{7}, \frac{5}{7}) = \emptyset$: no $\xi > 0$ has all of $\{\xi(3/2)^n\}$ in $[\frac{2}{7}, \frac{5}{7})$, a window of length $\frac{3}{7} = 0.428571\dots$. Moreover, for every $\xi \neq 0$ and every N there is $n \geq N$ with $\{\xi(3/2)^n\} \notin [\frac{2}{7}, \frac{5}{7}]$; equivalently,

$$\|\xi(3/2)^n\| < \frac{2}{7} = 0.285714\dots \quad \text{for infinitely many } n.$$

The certificate has funnel depth 15, 166 blocks in 32 rank strata, over the denominator $7 \cdot 3^{15}$. The window passes the corpus record 0.40722 and [Dub19, Thm. 1.2]’s $\frac{31}{81} = 0.38271\dots$, the longest single window excluded in print; the nearest-integer constant $2/7$ improves the corpus’s previous $\frac{1}{3}$, which came from the length- $\frac{1}{3}$ line at the centred window.²

7.1. The exact ceiling is a Thue–Morse constant. Write

$$T(z) := \prod_{m \geq 0} (1 - z^{2^m}),$$

as in [Dub06, (1)]. Corollary 1 of that paper states that for every $\xi \neq 0$ the sequence $(\|\xi(3/2)^n\|)_n$ has a limit point at least $(3 - T(2/3))/12 = 0.238117\dots$ and a limit point at most $(1 + T(2/3))/4 = 0.285647\dots$.

Theorem 7.1 misses the second of these constants by $6.7 \cdot 10^{-5}$, and [Ste26, Thm. 7.14] — which gives $\|\xi(3/2)^n\| \geq 0.238$ for infinitely many n — misses the first by $1.2 \cdot 10^{-4}$. Neither gap will close, and the reason is the same on both sides.

On the centred line write the window as $[c, 1 - c)$; an orbit of the model relation lies inside exactly when $c \leq \min_i \|y_i\|$. Sorting the periodic orbits by that minimum gives the order in which they enter the window as c falls, and the answer is a period-doubling cascade.

²The centred axis $s = (1 - L)/2$ is an axis of symmetry of the certified region, by an identity outside the formalization: the reflection $y \mapsto 1 - y$ conjugates the model relation to itself,

$$1 - \frac{py - s}{q} = \frac{p(1 - y) - (p - q - s)}{q},$$

so the carry is permuted by $s \mapsto p - q - s$ — at the base $3/2$ by $s \mapsto 1 - s$ — up to the half-open convention at the endpoints.

Conjecture 7.2. *At the base $3/2$ the orbits entering the centred window have periods 2^k and carry words the Thue–Morse prefixes of the same length, up to a cyclic shift, and their entry points c_k satisfy*

$$4c_k - 1 = \frac{3 \prod_{n=1}^{k-2} (3^{2^n} - 2^{2^n})}{3^{2^{k-1}} + 2^{2^{k-1}}}, \quad c_1 = \frac{2}{5}, \quad c_2 = \frac{4}{13}, \quad c_3 = \frac{28}{97}, \quad c_4 = \frac{1948}{6817}, \quad \dots$$

so that $c_k \downarrow (1 + T(2/3))/4$.

This statement is not formalized: it is the outcome of an exact rational computation, verified against a complete enumeration of the periodic orbits of period at most 20, and the closed form is checked for $k \leq 4$. The limit is immediate from the closed form: dividing numerator and denominator by powers of 3 leaves $\frac{1}{3} \prod_{n \geq 1} (1 - (2/3)^{2^n}) = T(2/3)$.

Granting Conjecture 7.2, every $c < (1 + T(2/3))/4$ puts all the c_k inside the window — infinitely many periodic orbits, whose points have denominator dividing $3^P - 2^P$ and so coprime to 6 — so Theorem 5.8 applies and the family stops exactly at Dubickas’s constant. That is what the measurements show: the family certifies the centred window at $c = 0.2856475$, only $1.8 \cdot 10^{-7}$ above the constant, and fails at $c = 0.2856472$. Read from the other side the same cascade is the one behind the two-interval entry of [Ste26], with entry points $0, \frac{1}{5}, \frac{3}{13}, \frac{23}{97}, \dots$ increasing to $(3 - T(2/3))/12$, measured there inside $(0.2381175, 0.2381177]$.

So both ceilings of this corpus are the two constants of [Dub06, Cor. 1], and the certificate family attains neither, because the printed constants are the accumulation points of its own obstruction. How much better a different method could do is a separate question, and one that is wide open: [Pol81] exhibits $\xi \neq 0$ with $\|\xi(3/2)^n\| > \frac{4}{65}$ for every n , and [AFS08] exhibits $\xi \neq 0$ with $\|\xi(3/2)^n\| < \frac{1}{3}$ for every n , so the two optimal constants lie in $[\frac{4}{65}, 0.285647\dots]$ and $[0.238117\dots, \frac{1}{3}]$ respectively.

7.2. What cuts the frontier row. It is not the count of periodic orbits in a window that decides, but the transient tree above them.

Remark 7.3. At the base $3/2$ one has $\frac{4}{15} \mapsto \frac{2}{5}$ and $\frac{11}{15} \mapsto \frac{3}{5}$ under the model relation, so a window containing $\frac{4}{15}$ traps an infinite backward chain accumulating on the 2-cycle $\{\frac{2}{5}, \frac{3}{5}\}$: the count of surviving funnel components is exactly $k + 1$ at depth k , and no finite partition into intervals ever separates the chain from the cycle, so the rank-certified positions of the row are exactly the windows whose closure lies in $(\frac{4}{15}, \frac{11}{15})$. Emptiness is not in doubt at the positions this refuses: every hold orbit there reaches the 2-cycle in finitely many steps, so Lemma 2.1 applies, and it is the certificate format, not the mathematics, that runs out. The same mechanism is the one [Ste26] isolated in [Dub19]’s own window, where the left endpoint $\frac{8}{57}$ is a preimage of a 3-cycle and moving it one grid unit inwards yields a certificate; it is not a curiosity of one printed theorem, but the only thing that cuts this row.

8. ACKNOWLEDGEMENTS

The author utilized Claude Code as an AI coding assistant to aid in the Lean 4 formalization of the proofs presented in this paper. The author directed and reviewed all generated code and takes full responsibility for the mathematical integrity and final content of the work.

APPENDIX A. FORMALIZATION

Every numbered statement above is a theorem of Lean 4 — or, where the table says so, an immediate arithmetic consequence of one — except the two marked “not formalized” in the tables below. The files live under Z32/ of the repository at

<https://github.com/rwst/Confinement-Certificates>, alongside the development of [Ste26]. “std3” abbreviates the ambient axioms of classical Lean (propositional extensionality, choice, quotient soundness). *Every* entry below is fully proved at std3: no **sorry**, no cited axioms, no natively compiled decision procedures. Rows marked “computed” are additionally verified by evaluating concrete integer data during type checking; the absence of that mark in §3 is the point of that section, whose proofs are uniform in (p, q, s, K) with no computation on concrete data. All footprints were confirmed with **#print axioms**. Two names predate the wording used above: **certifiedK** is Definition 3.3’s *escapes at depth K*, and **two_cell** is the two-interval union $[0, \frac{1}{5}) \cup [\frac{4}{5}, 1)$.

The schema and the rotation (§3). SymbolicCert (SC), DepthKSchema (DK).

Statement	Lean identifier	File	Status
Def. 3.1	Z32.schemaTheta, Z32.schemaBase, Z32.schemaEps	SC	def
Lem. 3.2	Z32.carry_eq_base_or_succ	SC	std3
the two branches	Z32.base_block, Z32.succ_block	SC	std3
Def. 3.3	Z32.Escape, Z32.SchemaCertifiedK	DK	def
r, N_L, N_R	Z32.blockRank, Z32.lowCount, Z32.highCount	DK	def
Prop. 3.4	Z32.blockRank_step_low, ..., step_high, Z32.low_iff_blockRank_le	DK	std3
Thm. 3.5	Z32.ZSet_eq_empty_of_certifiedK	DK	std3
core form	Z32.not_confined_of_certifiedK	DK	std3
branch = carry	Z32.step_of_confined, Z32.step_of_confined_at	DK	std3
Thm. 3.6	Z32.LowBand, Z32.escape_lowOrbit, Z32.ZSet_eq_empty_of_lowBand	DK	std3
closed form of w_i	Z32.lowOrbit, Z32.lowOrbit_add_eq	DK	std3
Cor. 3.7	Z32.SchemaCertified, Z32.ZSet_eq_empty_of_certified	SC	std3
clauses two, three = $K \leq 1$	Z32.certifiedK_of_certified	DK	std3
clause one, $\varepsilon = 0$	Z32.isEventuallyPeriodic_carry_of_certified, Z32.carry_const_base, ..., const_succ, ..., succ_of_base, ..., base_of_succ	SC	std3
core / i.o. forms	Z32.not_confined_of_certified, Z32.exists_fractions_of_certified	SC	std3
Cor. 3.8	—	—	arithmetic from (5)
Cor. 3.9(i)	Z32.ZSet_zero_eq_empty, Z32.ZSet_eq_empty_of_theta_int	SC	std3
Cor. 3.9(ii)	Z32.ZSet_half_eq_empty	SC	std3
Thm. 3.10	Z32.ZSet_five_two, ..., seven_two, ..., nine_two, ..., ten_three, ..., eleven_three	SC	std3
Cor. 3.11	Z32.ZSet_five_two_upper, ..., band, ..., interval	SC	std3
depths two, three	Z32.ZSet_five_two_depth_two, ..., depth_three, ..., depth_two_interval	DK	std3

The escape rate (§4).

Statement	Lean identifier	File	Status
M_n	Z32.orbDiff	EscapeBound	def
Def. 4.1	Z32.escapeSteps	EscapeBound	def
Thm. 4.2	Z32.not_period_of_pow_le, Z32.not_period_of_escapeSteps	EscapeBound	std3
<i>ladder</i>	Z32.q.pow.mul_orbDiff, Z32.q.pow.dvd_orbDiff	EscapeBound	std3
<i>dichotomy</i> (6)	Z32.escape_endgame	EscapeBound	std3
Prop. 4.3(i)	Z32.exists_carry_period_of_certifiedK	EscapeBound	std3
Prop. 4.3(ii)	Z32.BlockCert.Cert.exists_carry_period_le	EscapeBound	std3
Thm. 4.4, row two	Z32.BlockCert.Cert.exists_escape_le	EscapeBound	std3
<i>row one</i>	Z32.exists_escape_le_of_certifiedK, Z32.exists_fract_notMem_le_of_certifiedK	EscapeBound	std3
<i>covering form</i>	Z32.BlockCert.exists_fract_notMem_le_of_cover, ..._of_cover.pow	EscapeBound	std3
Tab. 1, rows 1–5, 7–9	Z32.BlockCert.escape_sixth_3.8, ..._frontier, ..._union_712, ..._union_23, ..._union_2536, ..._two_cell_fifth, ..._four_three, ..._five_two_fifth	EscapeBound	computed
Tab. 1, row 6	Z32.BlockCert.escape_union_7083	UnionRecord	computed
Cor. 4.5	Z32.BlockCert.escape_sixth_3.8_million	EscapeBound	computed

Completeness and the obstruction (§5).

Statement	Lean identifier	File	Status
Def. 5.1	Z32.holdSet	CertComplete	def
denominator coprime to q	Z32.HasDenom, Z32.CoprimeDenom	CertComplete	def
Thm. 5.2	Z32.step_unique	CertComplete	std3
Prop. 5.3	Z32.hasDenom_of_return, Z32.cyclePoint_eq_base	CertComplete	std3
<i>coprimality</i>	Z32.cycleDenom, Z32.cycleDenom_coprime	CertComplete	std3
Prop. 5.4	Z32.cycOrbit, Z32.cycOrbit_rec	CycleTransversal	std3
(i) <i>no transients</i>	Z32.cycOrbit_periodic, Z32.exists_periodic_orbit	CycleTransversal	std3
(ii) <i>converse</i>	Z32.cycOrbit_period_dvd	CycleTransversal	std3
<i>the P-periodic points</i>	Z32.cyclePoint_eq_base with Z32.cycOrbit_period_dvd	both	std3
Thm. 5.5	Z32.holdSet_finite_imp	CertComplete	std3
<i>finite range</i>	Z32.isEventuallyPeriodic_of_finite_range	CertComplete	std3
Thm. 5.6	Z32.BlockCert.Cert.eq_of_memI_block	CertComplete	std3
<i>block out-degree</i>	Z32.BlockCert.outEdges_length_le_one	CertComplete	std3
Thm. 5.7	Z32.BlockCert.Cert.ok_eq_false_of_infinite_hold	CertComplete	std3
Thm. 5.8	Z32.BlockCert.Cert.ok_eq_false_of_infinite_cycles	CycleTransversal	std3
Cor. 5.9	Z32.BlockCert.Cert.ok_eq_false_of_two_cell_chain, ..._ok_eq_false_of_covers_two_cell	CertComplete	std3
<i>the backward chain</i>	Z32.twoCellPt, Z32.twoCellOrbit, Z32.twoCellOrbit_rec	CertComplete	std3
Thm. 5.10	Z32.BlockCert.two_cell_fifth_closed, ..._sixth_3.8_closed, ..._frontier_closed, ..._four_three_closed, ..._five_two_fifth_closed	CertComplete	computed
<i>the five certificates</i>	Z32.BlockCert.certTwoCellFifthClosed, ..._certWindow38Closed, ..._certFrontierClosed, ..._certFourThreeClosed, ..._certFiveTwoClosed	CertComplete	computed
the 3-cycle of Rem. 6.6	Z32.carry_cycle_nineteen	CycleTransversal	std3

The depth-size bound (§6) — file *DepthSize*.

Statement	Lean identifier	Status
f^{-1} , slice, fold, T_k	Z32.pre, Z32.slice, Z32.fold, Z32.funnel	def
<i>factorisation</i>	Z32.pre_eq_slice_fold, Z32.subset_slice_fold, Z32.volume_slice	std3
Prop. 6.1	Z32.volume_le_volume_pre, Z32.one_le_volume_funnel_add	std3
Prop. 6.2	Z32.BlockCert.abs_sub_le_of_memI_block, ..._volume_funnel_le_of_cert	std3
Thm. 6.3	Z32.depth_size, Z32.depth_size_logb, Z32.BlockCert.cert_depth_size	std3
Cor. 6.4	Z32.BlockCert.volume_hole_pos, ..._volume_certSet_lt_one	std3
Ex. 6.5	Z32.BlockCert.hole_union_2536	computed

The ranked frontier (§7) — file *RankedFrontier*.

Statement	Lean identifier	Status
the certificate	Z32.BlockCert.certTwoSeven, ..._certTwoSeven_ok	computed
Thm. 7.1	Z32.ZSet.three.two.two_seven, Z32.two_seven_empty	computed
<i>eventual form</i>	Z32.not_eventually_two_seven	computed
<i>nearest-integer form</i>	Z32.not_forall_two_seven_le_abs_sub_round	computed
Conj. 7.2, Rem. 7.3	—	not formalized

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